1/1 Internal Kink Mode and Kadomtsev Reconnection Model in Tokamak Plasmas

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I. EQUILIBRIUM PROFILE

Let a, R_0, B_0, q_0, q_a, p_0, and ϵ_a be the minor and major radii, toroidal magnetic field, central safety factor, edge safety factor, central pressure, and inverse aspect ratio a/R_0, respectively. We define the normalized radius $\tilde{r} = r/a$ with $\tilde{r} \in [0, 1]$ and $\nu = q_a/q_0$. The equilibrium profile before the sawtooth crash $q_i(\tilde{r})$ is given by

$$q_i(\tilde{r}) = q_0 + (q_a - q_0)\tilde{r}^2 \tag{1}$$

with $q_i(\tilde{r} = 0) = q_0$, $q_i(\tilde{r} = 1) = q_a$; the pressure profile before the crash $p_i(\tilde{r})$ is given by

$$p_i(\tilde{r}) = p_0(1 - \tilde{r}^2)^\nu, \tag{2}$$

and the poloidal magnetic field $B_\theta(\tilde{r})$ is given by

$$B_\theta(\tilde{r}) = \frac{\tilde{r}B_0\epsilon_a}{q_i(\tilde{r})}. \tag{3}$$

In a tokamak, the magnetic field is

$$\mathbf{B} = B_\theta(\tilde{r})\hat{\boldsymbol{\theta}} + B_\phi(\tilde{r}, \theta)\hat{\boldsymbol{\phi}} \tag{4}$$

where $B_\phi(\tilde{r}, \theta) \approx B_0$ for large aspect ratio tokamaks with $r/R_0 \ll 1$. For the $m/n = 1/1$ mode, at $q = 1$, the poloidal field satisfies $B_\theta/B_0 = a\tilde{r}/R_0$, and we define the effective field

$$\mathbf{B}^*(\tilde{r}) = \mathbf{B} - \frac{B_0 r}{R_0}\hat{\boldsymbol{\theta}} = \Big(B_\theta(\tilde{r}) - q(\tilde{r})B_\theta(\tilde{r})\Big)\hat{\boldsymbol{\theta}} + B_0\hat{\boldsymbol{\phi}} \tag{5}$$

where

$$q(\tilde{r}) = \frac{\tilde{r}B_0\epsilon_a}{B_\theta(\tilde{r})} \tag{6}$$

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from Eq. 3.

We first study the poloidal component of the effective magnetic field $B^*_\theta(\tilde{r}) = B_\theta(\tilde{r})[1 - q(\tilde{r})]$ by introducing the effective poloidal flux $\psi^*(\tilde{r})$ with the given relation

$$\frac{\mathrm{d}\psi^*}{\mathrm{d}\tilde{r}} = aB^*_\theta(\tilde{r}) = aB_\theta(\tilde{r})[1 - q(\tilde{r})], \tag{7}$$

where such flux satisfies $\mathbf{B}^* = \hat{\boldsymbol{\phi}} \times \nabla\psi^*$ (following the right-handed orientation). The effective poloidal flux is obtained by

$$\psi^*(\tilde{r}) = a\int_0^{\tilde{r}} \frac{\mathrm{d}\psi^*}{\mathrm{d}\tilde{r}}\mathrm{d}\tilde{r} = a\int_0^{\tilde{r}} \left[B_\theta(\tilde{r}') - \tilde{r}' B_0 \epsilon_a\right]\mathrm{d}\tilde{r}. \tag{8}$$

The flux may be normalized as well

We account for the temperature and resistivity profile, where $T(\tilde{r})$ and $\eta(T)$ are given by

$$T(\tilde{r}) = T_c(1 - \tilde{r}^2)^\nu \tag{9}$$

and the Spitzer resistivity model

$$\eta(T) = \eta_0 T^{-3/2}, \tag{10}$$

with temperature and resistivity parameters T_c and η_0.

II. SAWTOOTH OSCILLATION AND KADOMTSEV RECONNECTION

The sawtooth instability occurs as the central safety factor $q(0)$ falls below 1; in other words, the crash is a characteristic internal kink mode on the $m/n = 1/1$ resonant surface. This instability exhibits a series of periodic 'sawtooth' drops in the plasma current, temperature, and pressure inside the instability boundary radius, shifting the magnetic axis. We are concerned with the ideal 1/1 kink mode scenario with the initial conditions $q_0 < 1$ and $q_a > 1$.

We numerically evaluate the radial coordinates as the instability occurs according to the model developed above. Consider

$$\Delta\tilde{r} = \frac{1 - \tilde{r}_{\min}}{N_r - 1} \tag{11}$$

with the initial offset $\tilde{r}_{\min}$,

$$\tilde{r}_j = \tilde{r}_{\min} + j\,(\Delta\tilde{r}), \tag{12}$$

$$q_i(\tilde{r}_j) = q_0 + (q_a - q_0)\tilde{r}_j^2, \tag{13}$$

$$B_\theta(\tilde{r}_j) = \frac{\tilde{r}_j B_0 \epsilon_a}{q_i(\tilde{r}_j)}, \tag{14}$$

and

$$\left[\frac{\mathrm{d}\psi^*}{\mathrm{d}\tilde{r}}\right]_j = aB_\theta(\tilde{r}_j)[1 - q_i(\tilde{r}_j)]. \tag{15}$$

The effective flux ψ_j^* is calculated by

$$\psi^*(\tilde{r}_j) = \psi^*(\tilde{r}_{j-1}) + \frac{\Delta\tilde{r}}{2}\left(\left[\frac{\mathrm{d}\psi^*}{\mathrm{d}\tilde{r}}\right]_j + \left[\frac{\mathrm{d}\psi^*}{\mathrm{d}\tilde{r}}\right]_{j-1}\right) \tag{16}$$

with the initial condition

$$\psi^*(\tilde{r}_{\min}) = 0. \tag{17}$$

To find the normalized resonance radius $\tilde{r}_1$ where the instability occurs; i.e., $\tilde{r}$ such that $B_\theta^*(\tilde{r}) = 0$, we evaluate $\tilde{r}_1$ where

$$q(\tilde{r}) - 1 = 0. \tag{18}$$

$\tilde{r}_1$ is found by the bisection method over $[\tilde{r}_{\min}, \tilde{r}_{N_r-1}]$. Consequently, we evaluate the boundary radius $\tilde{r}_0$ by considering

$$\psi^*(\tilde{r}) - \psi^*(0) = 0, \tag{19}$$

similarly by applying bisection to find the root over $[\tilde{r}_1 + \delta\tilde{r}_1, 1 - \delta\tilde{r}_a]$, where $\delta\tilde{r}_1$ and $\delta\tilde{r}_a$ are arbitrary offsets.

As $\tilde{r}_1$ and $\tilde{r}_0$ are obtained, $\tilde{r}_-$ and $\tilde{r}_+$ are evaluated such that

$$\tilde{r}_- < \tilde{r}_1 < \tilde{r}_+ < \tilde{r}_0 \tag{20}$$

with

$$\tilde{r}_- \in [0, \tilde{r}_1), \quad \tilde{r}_+ \in (\tilde{r}_1, \tilde{r}_0), \tag{21}$$

and the reconnection of field lines is considered, which follows

$$\psi^*(\tilde{r}_-) = \psi^*(\tilde{r}_+). \tag{22}$$

$\tilde{r}_-$ is computed using

$$\tilde{r}_{-,n} = \frac{\tilde{r}_1(n + \frac{1}{2})}{N'} \tag{23}$$

with index N' determining the grid, and $\tilde{r}_+$ is consequently found by bisection on $[\tilde{r}_1, \tilde{r}_0]$.

The condition

$$\tilde{r}_f^2 = \int_{\tilde{r}_1}^{\tilde{r}_+} \mathrm{d}\tilde{r}_+^2 + \int_{\tilde{r}_-}^{\tilde{r}_1} \mathrm{d}\tilde{r}_-^2 = \tilde{r}_+^2 - \tilde{r}_-^2 \tag{24}$$

is imposed for the post-crash profile, conserving the toroidal flux as two flux surfaces reconnect. The flux surfaces are paired following

$$\kappa_r \equiv \left| \frac{\mathrm{d}\tilde{r}_-}{\mathrm{d}\tilde{r}_+} \right| = \left| \frac{\mathrm{d}\psi^*}{\mathrm{d}\tilde{r}}\Big|_{\tilde{r}_+} \right| \cdot \left| \frac{\mathrm{d}\psi^*}{\mathrm{d}\tilde{r}}\Big|_{\tilde{r}_-} \right|^{-1}, \tag{25}$$

with the final temperature after flux pairing

$$T_f = \frac{T(\tilde{r}_+)\tilde{r}_+ + T(\tilde{r}_-)\kappa_r \tilde{r}_-}{\tilde{r}_+ + \kappa_r \tilde{r}_-}. \tag{26}$$

The temperature relaxation is understood to be immediate, thus we neglect the relaxation timescale and approximate it as 0, effectively. As $\frac{\mathrm{d}\psi^*}{\mathrm{d}\tilde{r}}\big|_{\tilde{r}_-} < \delta$ for a small parameter δ is satisfied, $\frac{\mathrm{d}\tilde{r}_-}{\mathrm{d}\tilde{r}_+}$ is set to 1. Once the sawtooth crash completes, T, q, and η are recorded and B_θ is computed. From the recorded q, the flux ψ^* is calculated. The post-crash q profile is given by

$$q_f(\tilde{r}) = \begin{cases} 1 + \big(q_i(\tilde{r}_0) - 1\big)\big(\frac{\tilde{r}}{\tilde{r}_0}\big)^2, & \tilde{r} \le \tilde{r}_0 \\ q_i(\tilde{r}), & \tilde{r} > \tilde{r}_0 \end{cases} \tag{27}$$

and consequently, the poloidal magnetic field is

$$B_{\theta,f} = \frac{\tilde{r} B_0 \epsilon_a}{q_f}, \tag{28}$$

with q_i and q_f indicating the initial and final q profiles.

During the current ramp, q undergoes relaxation with a rather rapid relaxation timescale τ_R, and as $q(0)$ falls below 1, the sawtooth crash is triggered again.

The time evolutions of the temperature profile is given by

$$\frac{3}{2} n_e e \frac{\partial T}{\partial t} + \nabla \cdot (-n_e \chi \nabla T) = \eta J^2, \tag{29}$$

which yields

$$\frac{3}{2} n_e e \frac{\partial T}{\partial t} = \frac{1}{\tilde{r}} \frac{\partial}{\partial \tilde{r}} \left(\tilde{r} \tilde{\chi} n_e \frac{\partial T}{\partial \tilde{r}} \right) + \eta J^2 \tag{30}$$

with the normalized diffusion coefficient $\tilde{\chi} = \chi / a^2$ and electron number density n_e; the current density $J = J_z$ is given by

$$J = \frac{1}{\mu_0 a \tilde{r}} \frac{\mathrm{d}(\tilde{r} B_\theta)}{\mathrm{d}\tilde{r}} = \frac{B_0 \epsilon_a}{\mu_0 a q} \left(2 - \frac{\tilde{r}}{q} \frac{\mathrm{d}q}{\mathrm{d}\tilde{r}} \right). \tag{31}$$

The time evolution of q is given by

$$\frac{\partial q}{\partial t} = \frac{(q_0 + (q_a - q_0)\tilde{r}^2) - q(\tilde{r})}{\tau_R} \tag{32}$$

as q relaxes back to the initial condition.

It seems sufficient to use Euler's method to obtain T for a small timestep Δt with large enough diffusion timescale relative to Δt; the radial diffusion is obtained via backward Euler scheme by solving the tridiagonal matrix using the Thomas algorithm, given the boundary conditions

$$\left.\frac{\partial T}{\partial \tilde{r}}\right|_{\tilde{r}=0} = 0, \quad T(\tilde{r}=1) = T_a \tag{33}$$

for the edge temperature T_a. The ohmic heating term ηJ^2 may be treated using the forward Euler scheme, as well as the safety factor q.